

20. (g) Implementation Plan: B.Tech. in Computer Science and Engineering with Specialization in Big Data Analytics

| Semester – I           |                                        |              |   |   |   | Semester – II                                                                                  |                                                                            |              |   |    |    |
|------------------------|----------------------------------------|--------------|---|---|---|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------|---|----|----|
| Course Code            | Course Title                           | Hours / Week |   |   | C | Course Code                                                                                    | Course Title                                                               | Hours / Week |   |    | C  |
|                        |                                        | L            | T | P |   |                                                                                                |                                                                            | L            | T | P  |    |
| 21LEH102T              | Chinese                                | 2            | 1 | 0 | 3 | 21LEH101T                                                                                      | Communicative English                                                      | 2            | 1 | 0  | 3  |
| 21LEH103T              | French                                 |              |   |   |   | 21MAB102T                                                                                      | Advanced Calculus and Complex Analysis                                     | 3            | 1 | 0  | 4  |
| 21LEH104T              | German                                 |              |   |   |   | 21PYB102J                                                                                      | Semiconductor Physics and Computational Methods                            | 3            | 1 | 2  | 5  |
| 21LEH105T              | Japanese                               |              |   |   |   | 21MES102L <sup>1</sup>                                                                         | Engineering Graphics and Design                                            | 0            | 0 | 4  | 2  |
| 21LEH106T              | Korean                                 |              |   |   |   | 21EES101T                                                                                      | Electrical and Electronics Engineering                                     | 3            | 1 | 0  | 4  |
| 21LEH107T              | Spanish                                |              |   |   |   | 21CSC101T                                                                                      | Object Oriented Design and Programming                                     | 2            | 1 | 0  | 3  |
| 21LEH108T              | Russian                                |              |   |   |   | 21CYM101T <sup>1</sup>                                                                         | Environmental Science*                                                     | 1            | 0 | 0  | 0  |
| 21GNH101J              | Philosophy of Engineering              | 1            | 0 | 2 | 2 | 21PDM102L <sup>1</sup>                                                                         | General Aptitude*                                                          | 0            | 0 | 2  | 0  |
| 21MAB101T              | Calculus and Linear Algebra            | 3            | 1 | 0 | 4 | 21LEM101T <sup>1</sup>                                                                         | Constitution of India                                                      | 1            | 0 | 0  | 0  |
| 21CYB101J              | Chemistry                              | 3            | 1 | 2 | 5 | Total Credits 21                                                                               |                                                                            |              |   |    |    |
| 21BTB102T              | Introduction to Computational Biology  | 2            | 0 | 0 | 2 | Semester – IV                                                                                  |                                                                            |              |   |    |    |
| 21CSS101J              | Programming for Problem Solving        | 3            | 0 | 2 | 4 | Course Code                                                                                    | Course Title                                                               | Hours / Week |   |    | C  |
| 21MES101L <sup>1</sup> | Basic Civil and Mechanical Workshop    | 0            | 0 | 4 | 2 |                                                                                                |                                                                            | L            | T | P  |    |
| 21PDM101L <sup>1</sup> | Professional Skills and Practices      | 0            | 0 | 2 | 0 | 21MAB301T                                                                                      | Probability and Statistics                                                 | 3            | 1 | 0  | 4  |
| 21GNM101L <sup>1</sup> | Physical and Mental Health using Yoga  | 0            | 0 | 2 | 0 | 21CSC204J                                                                                      | Design and Analysis of Algorithms                                          | 3            | 0 | 2  | 4  |
| 21GNM102L <sup>1</sup> | National Service Scheme                |              |   |   |   | 21CSC205P <sup>1</sup>                                                                         | Database Management Systems                                                | 3            | 1 | 0  | 4  |
| 21GNM103L <sup>1</sup> | National Cadet Corps                   |              |   |   |   | 21CSC206T                                                                                      | Artificial Intelligence                                                    | 2            | 1 | 0  | 3  |
| 21GNM104L <sup>1</sup> | National Sports Organization           |              |   |   |   | E                                                                                              | Professional Elective-I                                                    |              |   |    | 3  |
| Total Credits 22       |                                        |              |   |   |   | 21PDH209T <sup>1</sup>                                                                         | Social Engineering                                                         | 2            | 0 | 0  | 2  |
| Semester – III         |                                        |              |   |   |   | 21PDM202L <sup>1</sup>                                                                         | Critical and Creative Thinking Skills                                      | 0            | 0 | 2  | 0  |
| Course Code            | Course Title                           | Hours / Week |   |   | C | 21LEM202T <sup>1</sup>                                                                         | Universal Human Values-II: Understanding Harmony and Ethical Human Conduct | 2            | 1 | 0  | 3  |
|                        |                                        | L            | T | P |   | Total Credits 23                                                                               |                                                                            |              |   |    |    |
| 21MAB201T              | Transforms and Boundary Value Problems | 3            | 1 | 0 | 4 | Semester – VI                                                                                  |                                                                            |              |   |    |    |
| 21DCS201P <sup>1</sup> | Design Thinking and Methodology        | 1            | 2 | 0 | 3 | Course Code                                                                                    | Course Title                                                               | Hours / Week |   |    | C  |
| 21CSS202T              | Fundamentals of Data Science           | 3            | 2 | 0 | 5 |                                                                                                |                                                                            | L            | T | P  |    |
| 21CSC201J              | Data Structures and Algorithms         | 3            | 0 | 2 | 4 | 21CSS301T                                                                                      | Full Stack Development                                                     | 1            | 1 | 0  | 2  |
| 21CSC202J              | Operating Systems                      | 3            | 0 | 2 | 4 | 21CSC303J                                                                                      | Software Engineering and Project Management                                | 2            | 0 | 2  | 3  |
| 21CSC206P <sup>1</sup> | Advanced Object Oriented Programming   | 2            | 1 | 0 | 3 | 21CSC304J                                                                                      | Compiler Design                                                            | 2            | 0 | 2  | 3  |
| 21LEM201T <sup>1</sup> | Professional Ethics*                   | 1            | 0 | 0 | 0 | E                                                                                              | Professional Elective – III                                                |              |   |    | 3  |
| 21PDM201L <sup>1</sup> | Verbal Reasoning*                      | 0            | 0 | 2 | 0 | E                                                                                              | Professional Elective – IV                                                 |              |   |    | 3  |
| Total Credits 23       |                                        |              |   |   |   | O                                                                                              | Open Elective – II                                                         |              |   |    | 3  |
| Semester – V           |                                        |              |   |   |   | 21CSP302L <sup>1</sup>                                                                         | Project                                                                    | 0            | 0 | 6  | 3  |
| Course Code            | Course Title                           | Hours / Week |   |   | C | 21CSP303T <sup>1</sup>                                                                         | MOOC                                                                       | 3            | 0 | 0  | 0  |
|                        |                                        | L            | T | P |   | 21PDM302L <sup>1</sup>                                                                         | Employability Skills and Practices                                         | 0            | 0 | 2  | 0  |
| 21MAB302T              | Discrete Mathematics                   | 3            | 1 | 0 | 4 | 21LEM302T <sup>1</sup>                                                                         | Indian Traditional Knowledge                                               | 1            | 0 | 0  | 0  |
| 21CSC301T              | Formal Language and Automata           | 3            | 0 | 0 | 3 | Total Credits 20                                                                               |                                                                            |              |   |    |    |
| 21CSC302J              | Computer Networks                      | 3            | 0 | 2 | 4 | Semester - VIII                                                                                |                                                                            |              |   |    |    |
| 21CSC307P <sup>1</sup> | Machine Learning for Data Analytics    | 2            | 1 | 0 | 3 | Course Code                                                                                    | Course Title                                                               | Hours / Week |   |    | C  |
| E                      | Professional Elective – II             |              |   |   | 3 |                                                                                                |                                                                            | L            | T | P  |    |
| O                      | Open Elective – I                      |              |   |   | 3 | 21CSP401L                                                                                      | Major Project                                                              | 0            | 0 | 30 | 15 |
| 21GNP301L <sup>1</sup> | Community Connect                      | 0            | 0 | 2 | 1 | 21CSP402L                                                                                      | Major Project                                                              | 0            | 0 | 20 | 10 |
| 21PDM301L <sup>1</sup> | Analytical and Logical Thinking Skills | 0            | 0 | 2 | 0 | 21CSP403L                                                                                      | Internship#                                                                | 0            | 0 | 10 | 5  |
| 21LEM301T <sup>1</sup> | Indian Art Form                        | 1            | 0 | 0 | 0 | Total Credits 15                                                                               |                                                                            |              |   |    |    |
| Total Credits 21       |                                        |              |   |   |   | Semester – VII                                                                                 |                                                                            |              |   |    |    |
| Semester – VII         |                                        |              |   |   |   | Course Code                                                                                    | Course Title                                                               | Hours / Week |   |    | C  |
| Course Code            | Course Title                           | Hours / Week |   |   | C |                                                                                                |                                                                            | L            | T | P  |    |
|                        |                                        | L            | T | P |   | 21GNH401T                                                                                      | Behavioral Psychology                                                      | 2            | 1 | 0  | 3  |
| E                      | Professional Elective – V              |              |   |   | 3 | E                                                                                              | Professional Elective – VI                                                 |              |   |    | 3  |
| E                      | Professional Elective – VI             |              |   |   | 3 | E                                                                                              | Professional Elective – VII                                                |              |   |    | 3  |
| E                      | Professional Elective – VII            |              |   |   | 3 | E                                                                                              | Professional Elective – VIII                                               |              |   |    | 3  |
| O                      | Open Elective –III                     |              |   |   | 3 | Total Credits 18                                                                               |                                                                            |              |   |    |    |
| Total Credits 18       |                                        |              |   |   |   | #Students have to register either 21CSP401L or 21CSP402L and 21CSP403L both in eighth semester |                                                                            |              |   |    |    |

#Students have to register either 21CSP401L or 21CSP402L and 21CSP403L both in eighth semester